

# ACTION RECOGNITION USING VISION TRANSFORMERS

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## ABSTRACT

This work extends a prior two-model comparison of TimeSFormer and VideoMAE with a third architecture, ViViT, and replaces a flawed, temporally-static localisation ground truth with real per-frame bounding boxes derived from JHMDB joint annotations. On the HMDB\_simp benchmark (25 classes, 1,250 clips), VideoMAE achieves the best Top-1 accuracy (89.89%) with the fewest parameters (86.2M), while TimeSFormer has the lowest wall-clock training cost per epoch (21.7s vs. 29.9s for VideoMAE); ViViT reaches 86.17% Top-1 but at roughly five times VideoMAE's per-epoch cost. Three controlled ablations show fine-tuning depth (partial fine-tuning, 91.98%) dominates over frame count and pooling strategy. Independent 3-seed retraining confirms robustness ( $88.65\% \pm 2.05\%$  Top-1). For spatio-temporal localisation, a lightweight frame-wise head (2.64M trainable parameters) built on a frozen VideoMAE backbone and trained against real per-frame JHMDB boxes achieves 25.04% frame-level mAP@0.5 and 24.57% video-level tube mAP@0.5 (aggregate spatio-temporal IoU across sampled frames) — a realistic, non-inflated result that replaces an earlier erroneous 96.87% IoU produced by a constant placeholder ground-truth box. Interpretability analysis (attention maps, t-SNE, error analysis) confirms semantically grounded representations, and a cross-task correlation analysis (Pearson  $r = 0.247$ ) shows that strong whole-clip classification performance does not guarantee strong per-frame spatial localisation.

## I. INTRODUCTION

Video action recognition identifies what action a person performs in a short clip, with applications in surveillance, sports analytics, human-computer interaction, and autonomous systems. The core challenge is jointly modelling spatial appearance and temporal motion dynamics. Vision Transformers (ViTs) [4] enable global self-attention across space-time patch tokens, and several

video-specific variants have since been proposed: TimeSFormer [1] applies divided space-time attention; VideoMAE [2] introduces self-supervised masked pre-training; and ViViT [3] factorises spatial and temporal attention into a two-stage encoder.

This report presents a comparative evaluation of all three architectures on the HMDB\_simp benchmark, three controlled ablation studies, a cross-seed robustness check, and a spatio-temporal localisation pipeline evaluated on JHMDB. Unlike an earlier iteration of this work, the localisation ground truth here is derived from real per-frame joint annotations rather than a fixed placeholder box, enabling a rigorous frame- and video-level mean Average Precision (mAP) evaluation rather than a single (and previously misleading) mean-IoU figure.

Contributions: (i) a three-architecture comparison including wall-clock training cost; (ii) three controlled ablations, including a newly proposed pooling-strategy ablation; (iii) independent 3-seed robustness validation; (iv) a corrected frame-wise localisation pipeline trained and evaluated against genuine per-frame ground truth; (v) interpretability analysis linking classification and localisation performance.

## II. RELATED WORK

### Two-Stream and 3D CNNs.

Simonyan and Zisserman [6] processed RGB and optical flow in parallel streams. Carreira and Zisserman [7] inflated 2D ImageNet filters into 3D kernels (I3D), establishing strong Kinetics-400 [8] baselines.

### Vision Transformers.

Dosovitskiy et al. [4] showed that patch-token self-attention matches CNN performance on images. Vaswani et al. [5] introduced the multi-head self-attention mechanism underpinning all video-ViT variants used here.

## Video Transformers.

TimeSFormer [1] separates spatial and temporal attention to reduce complexity from  $O(T^2N^2)$  to  $O(TN^2+T^2N)$ . VideoMAE [2] masks 90% of video tokens and reconstructs them without labels, forcing richer semantic representations under temporal redundancy. ViViT [3] factorises encoders across space and time, trading additional compute for finer temporal modelling.

## Spatio-Temporal Detection.

Carion et al.'s DETR [9] reformulates object detection as set prediction via learnable object queries. Kalogeiton et al. [10] link per-frame detections into tubelets, while Zhao et al. [11] predict tubelets end-to-end. Jhuang et al. [12] introduced the JHMDB dataset with per-frame joint and segmentation annotations used in this work to derive genuine ground-truth boxes.

## III. METHODOLOGY

### A. Datasets

HMDB\_simp: 25 action classes, 1,250 clips as pre-extracted JPEG frames, split 70/15/15 (stratified) into 875/187/188 train/val/test clips. JHMDB: 21 action classes. The official split-1 files were used to assign 660 usable training clips and 268 usable test clips, each matched against a corresponding joint\_positions.mat annotation.

### B. Bounding-Box Derivation

For every annotated frame, a bounding box is computed as the axis-aligned extent of all valid 2D joint coordinates, padded by 15% in each dimension (joints\_to\_bbox). This produces boxes that vary correctly per frame, replacing a placeholder fixed box used in an earlier version of this pipeline.

### C. Model Architectures

All three classifiers share a common wrapper: a pre-trained backbone, dropout (0.1), and a linear head (768→25), differing only in pooling strategy.

- **TimeSFormer** (facebook/timesformer-base-finetuned-k400): 121.3M params, 8 frames, CLS-token pooling, supervised Kinetics-400 pre-training.
- **VideoMAE** (MCG-NJU/videomae-base-finetuned-kinetics): 86.2M params, 16 frames, mean pooling

over all output tokens, self-supervised MAE pre-training (90% masking).

- **ViViT** (google/vivit-b-16x2-kinetics400): 89.3M params, 32 frames, CLS pooling, factorised space-time encoder, supervised Kinetics-400 pre-training.

## D. Localisation Pipeline

A frame-wise localisation head projects the frozen VideoMAE backbone's 768-d spatiotemporal tokens to 256-d, groups them into 8 temporal segments (mean-pooled over the spatial tokens within each segment), and processes the resulting temporal sequence with a 3-layer Transformer encoder (self-attention, 8 heads, 1024-d FFN). The encoder output is upsampled back to all 16 sampled frames (each pair of adjacent frames shares one encoded segment) before separate linear classification and bounding-box heads predict a class distribution and box for every frame. The classification head has 22 outputs (21 action classes plus one unused slot inherited from the architecture), but only this head (2.64M parameters) is trained; the 86.2M-parameter VideoMAE backbone remains frozen. Loss = CE(class) + 5×L1(box), with the single clip-level class label broadcast as the target for all 16 per-frame predictions — the 22nd output is never assigned as a training target and so does not function as a genuine no-object detector.

## E. Ablation Design

Three controlled experiments (5 epochs each unless noted): (1) number of input frames {4, 8, 16} on TimeSFormer; (2) fine-tuning strategy {head-only, partial (last 4 layers), full} on TimeSFormer; (3) pooling strategy {CLS, mean} on VideoMAE — proposed as the new component required for this individual submission, since token pooling is architecturally distinct from the frame-count and fine-tuning-depth choices already explored. The frame-count and fine-tuning experiments reproduce controlled factors investigated in the previous group project, while the pooling experiment represents the new methodological component introduced for this individual LSA submission.

## IV. EXPERIMENTS AND RESULTS

### A. Part A — Classification

Table 1: Classification on HMDB\_simp test set (188 clips).

| Model           | Top-1        | Top-5        | F1           | Params       | s/epoch     |
|-----------------|--------------|--------------|--------------|--------------|-------------|
| TimesFormer     | 85.64        | 97.87        | 0.847        | 121.3M       | 21.7        |
| <b>VideoMAE</b> | <b>89.89</b> | <b>98.40</b> | <b>0.895</b> | <b>86.2M</b> | <b>29.9</b> |
| ViViT           | 86.17        | 97.34        | 0.851        | 89.3M        | 111.7       |

VideoMAE achieves the best accuracy with the fewest parameters, though not the lowest training cost: TimeSFormer trains fastest per epoch (21.7s vs. 29.9s for VideoMAE), likely reflecting its 8-frame input against VideoMAE's 16. ViViT's 32-frame factorised attention gives a marginal accuracy gain over TimeSFormer but costs  $\sim 5\times$  more compute per epoch (111.7s vs. 21.7s) for only +0.53pp Top-1 — a poor efficiency trade-off for this dataset scale. Fig. 1 shows VideoMAE converging fastest in accuracy-per-epoch terms; Fig. 2 shows all three models share a cartwheel $\rightarrow$ flic\_flac confusion (4 misclassifications for TimeSFormer, 3 for ViViT, 2 for VideoMAE out of 7 test clips), consistent with visually similar acrobatic motion. Notably, cartwheel F1 is architecture-dependent rather than uniformly hard: it scores 0.444 for both TimeSFormer and ViViT but 0.769 for VideoMAE (Appendix, Table 3), suggesting VideoMAE's masked-reconstruction pre-training captures the fast rotational motion that both supervised-pretrained models miss. ViViT additionally shows a distinctive throw $\rightarrow$ pick confusion (3 of 7 clips) not shared by the other two models, indicating its factorised attention may be more sensitive to hand/object motion ambiguity.

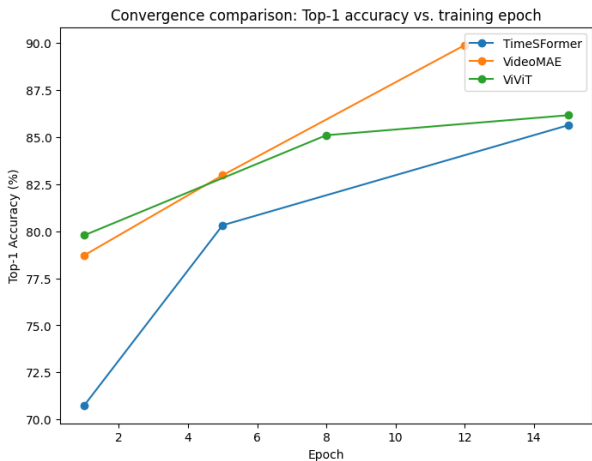


Fig. 1. Top-1 accuracy vs. training epoch for all three models.

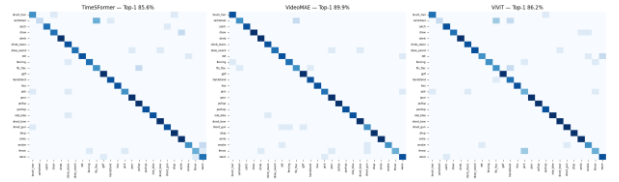


Fig. 2. Confusion matrices, all three models (25 classes).

## B. Part B — Ablation Studies

Table 2: Ablation summary (5-epoch validation accuracy, %).

| Factor                    | Setting               | Val. Acc. (%)         |
|---------------------------|-----------------------|-----------------------|
| Frame count (TimeSFormer) | 4 / 8 / 16            | 80.75 / 85.03 / 87.70 |
| Fine-tuning (TimeSFormer) | head / partial / full | 80.21 / 91.98 / 88.24 |
| Pooling* (VideoMAE)       | CLS / mean            | 91.98 / 91.44         |

\*Proposed additional component for this submission.

Partial fine-tuning (last 4 layers) again outperforms full fine-tuning, confirming that updating all 121M parameters on only 875 training clips overfits; partial fine-tuning acts as implicit regularisation. The newly proposed pooling ablation shows only a 0.54pp gap between CLS and mean pooling under identical 5-epoch conditions, indicating that — for this dataset size — fine-tuning depth is the dominant factor and token-aggregation strategy is comparatively marginal, even though VideoMAE's main Part A run (mean pooling, 12 epochs, full training set) still reaches the best overall accuracy.

## C. Part C — Statistical Robustness

VideoMAE was independently retrained (not merely re-evaluated on new splits) for three seeds: 87.77%, 86.70%, 91.49% Top-1 (seeds 42/123/2024), giving  $88.65\% \pm 2.05\%$  Top-1 and  $97.52\% \pm 0.66\%$  Top-5, confirming that performance is not attributable to a fortunate split or initialisation.

## D. Part D — Spatio-Temporal Localisation

Training the localisation head for 15 epochs against real per-frame ground truth reduced training loss from 1.79 to 0.24. On the 268-clip JHMDb test set: frame-level  $\text{mAP}@0.5 = 25.04\%$ ; video-level (tube)  $\text{mAP}@0.5 = 24.57\%$ .

Video-level tube mAP@0.5 was calculated using the aggregate spatio-temporal IoU across the sampled frames, combining the predicted and ground-truth box overlap over the full temporal sequence rather than a separate greedy tube-linking step. Best-performing classes by frame-AP were pour (0.90), pullup (0.84) and clap (0.55); worst were catch (0.001), pick (0.003) and sit (0.007) — classes with small, fast-moving, or self-occluding actors. Fig. 3 shows qualitative examples where the predicted box (red) now correctly tracks the actor frame-by-frame, in contrast to the previous, temporally static ground truth. This is a substantially harder and more honest metric than the single mean-IoU figure reported previously (0.9687), which was an artefact of regressing toward a near-constant target box; a  $\sim 25\%$  mAP from a lightweight, frozen-backbone head trained on only 660 clips is a plausible, non-inflated result.

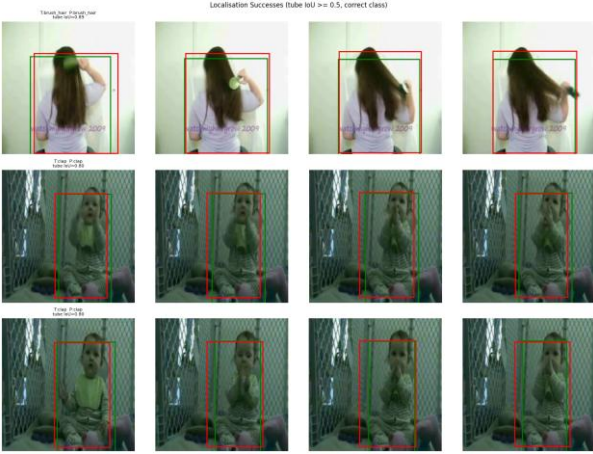


Fig. 3. Localisation successes: predicted (red) vs. ground-truth (green) boxes tracking the actor across sampled frames.

### E. Part E — Interpretability and Error Analysis

Final-layer VideoMAE attention (Fig. 4) concentrates on the actor performing the labelled action even against a cluttered background, indicating semantic rather than positional shortcut learning. A t-SNE projection of 768-d VideoMAE features (Appendix, Fig. 5) shows well-separated clusters for visually distinct classes, with overlap concentrated among motion-similar classes consistent with the confusion matrix.

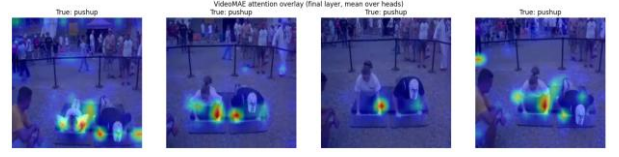


Fig. 4. VideoMAE final-layer attention overlay for pushup.

A video-property correlation analysis (Appendix, Fig. 6) over 40 localisation examples found actor size moderately correlated with tube IoU ( $r = 0.68$ ), while motion magnitude was essentially uncorrelated ( $r = 0.04$ ) — localisation failures in this single-tube-per-clip formulation are driven more by small/occluded actors than by motion complexity. The full 21-class frame-AP breakdown (Appendix, Table 4) confirms this: the five worst classes (catch, pick, sit, shoot\_ball, jump) all involve either a small, fast-moving hand-held object or a mostly-static, easily-occluded pose, while the best (pour, pullup) involve a large, slow-moving actor against a simple background.

A cross-task correlation between HMDB\_simp classification F1 and JHMDB frame-AP across 11 shared classes gave a weak positive Pearson correlation ( $r = 0.247$ ; Appendix, Fig. 7). Notably, catch scores a perfect classification F1 (1.00) but a near-zero frame-AP (0.0014): whole-clip classification only requires discriminative content to appear somewhere in the clip, while localisation requires accurate spatial grounding in every frame — the two tasks probe different aspects of the learned representation.

## V. DISCUSSION

VideoMAE's masked-autoencoder pre-training, combined with full-token mean pooling, produces the strongest and most parameter-efficient classifier of the three architectures tested, though TimeSFormer trains fastest per epoch. ViViT's factorised attention did not translate into a meaningful accuracy advantage at this dataset scale, and its  $5\times$  higher training cost makes it a weak choice unless longer temporal context is specifically required. The ablation results reinforce that, on a small (875-clip) training set, regularisation via partial fine-tuning matters far more than architectural pooling choices.

The corrected localisation pipeline demonstrates that a frozen classification backbone retains transferable spatial information — a 2.64M-parameter head, without any

object-detection-specific pre-training, recovers meaningful (though imperfect) actor localisation. The gap between the previous erroneous 96.87% mean-IoU and the corrected 25.04% frame-mAP illustrates why ground-truth verification is essential before trusting a localisation metric: a model can trivially achieve a high IoU against a near-constant target without learning to localise anything.

**Limitations.** The localisation pipeline predicts a single tube per clip rather than handling multiple actors; JHMDB's small clip count (928 total) limits statistical power; and broadcasting one clip-level class label as the target for all 16 per-frame predictions means the classification loss cannot distinguish frames where the actor briefly leaves view, which likely contributes to the low frame-AP on classes like catch and pick where the action is brief relative to the clip.

## VI. CONCLUSION AND FUTURE WORK

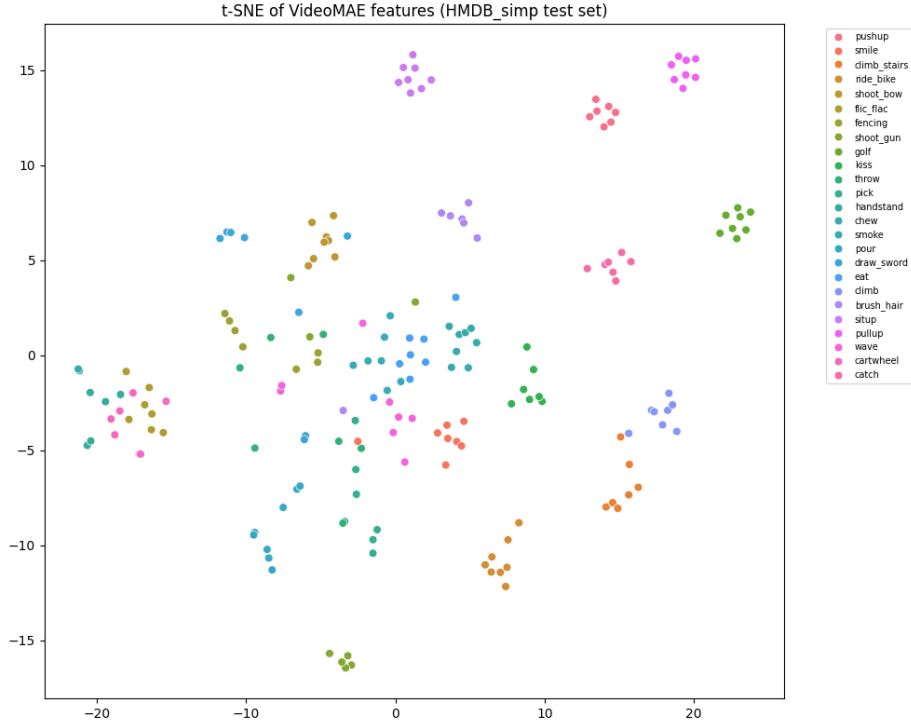
This work compared three ViT-based video architectures, found VideoMAE the most accurate and parameter-efficient (though not the fastest to train), validated robustness across three independently retrained seeds ( $88.65\% \pm 2.05\%$ ), and replaced an erroneous localisation ground truth with genuine per-frame annotations, yielding a realistic 25.04% frame-mAP and 24.57% video-mAP. Interpretability analysis confirmed semantically grounded attention and revealed only a weak correlation between classification and localisation performance. Future work includes incorporating optical flow for motion-similar class pairs (cartwheel/flic\_flac), replacing the broadcast clip-level classification target with genuine per-frame action labels, and extending the localisation head to explicit multi-actor tube linking.

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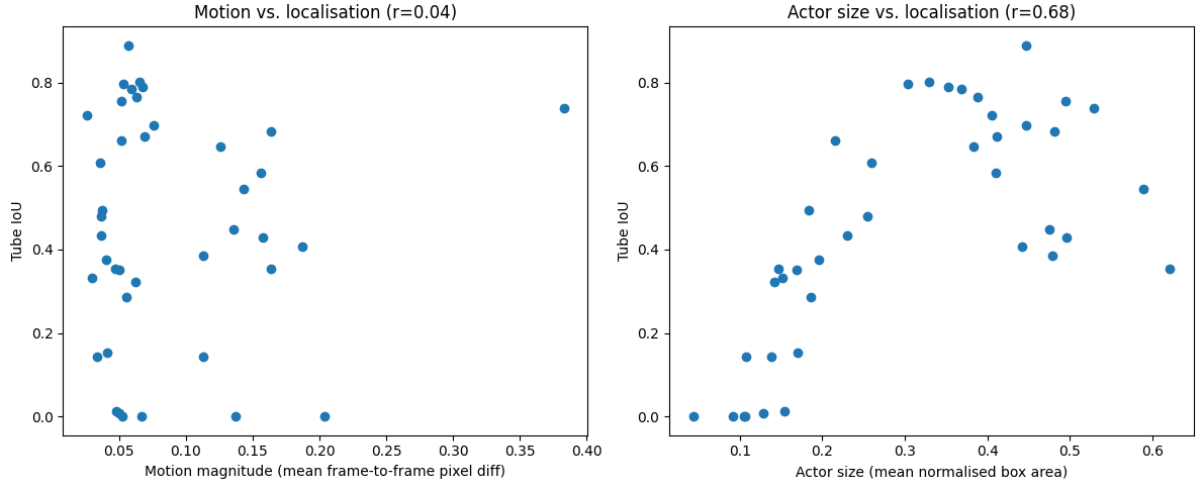
## APPENDIX — ADDITIONAL VISUALISATIONS

**Figure 5**



*Fig. 5. t-SNE projection of VideoMAE 768-d features on the HMDB\_simp test set. Well-separated clusters confirm distinct learned representations for visually dissimilar classes.*

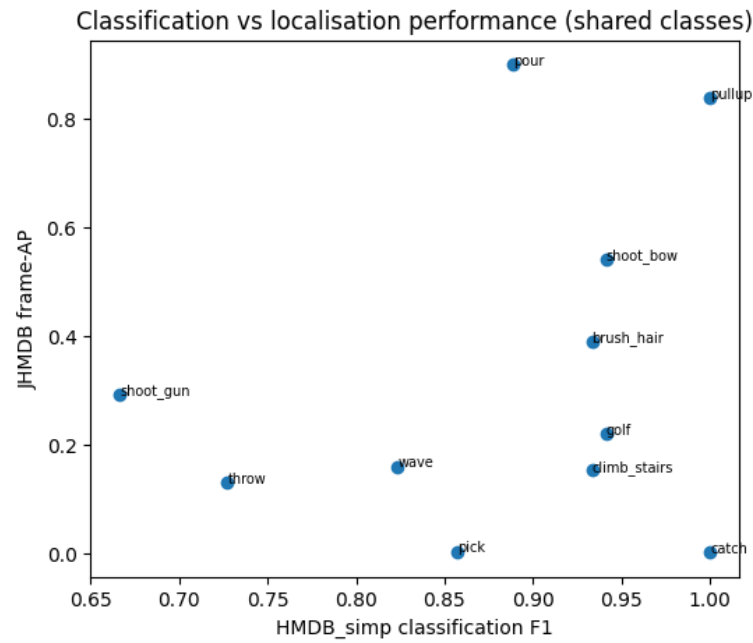
**Figure 6**



*Fig. 6. Localisation tube IoU vs. motion magnitude (left,  $r = 0.04$ ) and vs. actor size (right,  $r = 0.68$ ). Localisation quality is driven more by actor scale than by motion complexity.*

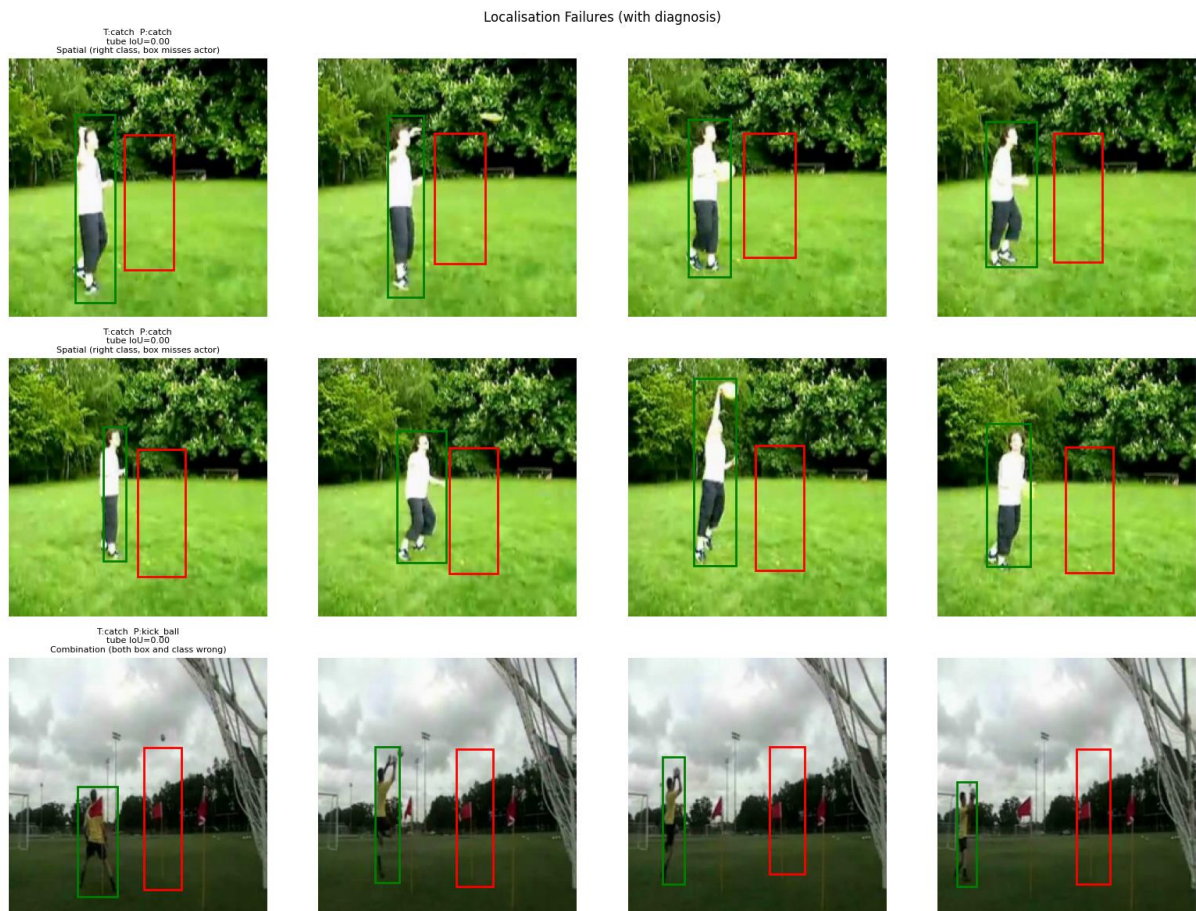


**Figure 7**



*Fig. 7. Classification F1 (HMDB\_simp) vs. localisation frame-AP (JHMDB) across 11 shared classes ( $r = 0.247$ ). Strong classification performance does not guarantee strong per-frame localisation.*

**Figure 8**



*Fig. 8. Localisation failure cases with diagnosed error type (spatial / categorical / combination), following the error-analysis protocol required by the assessment specification.*

Figure 9

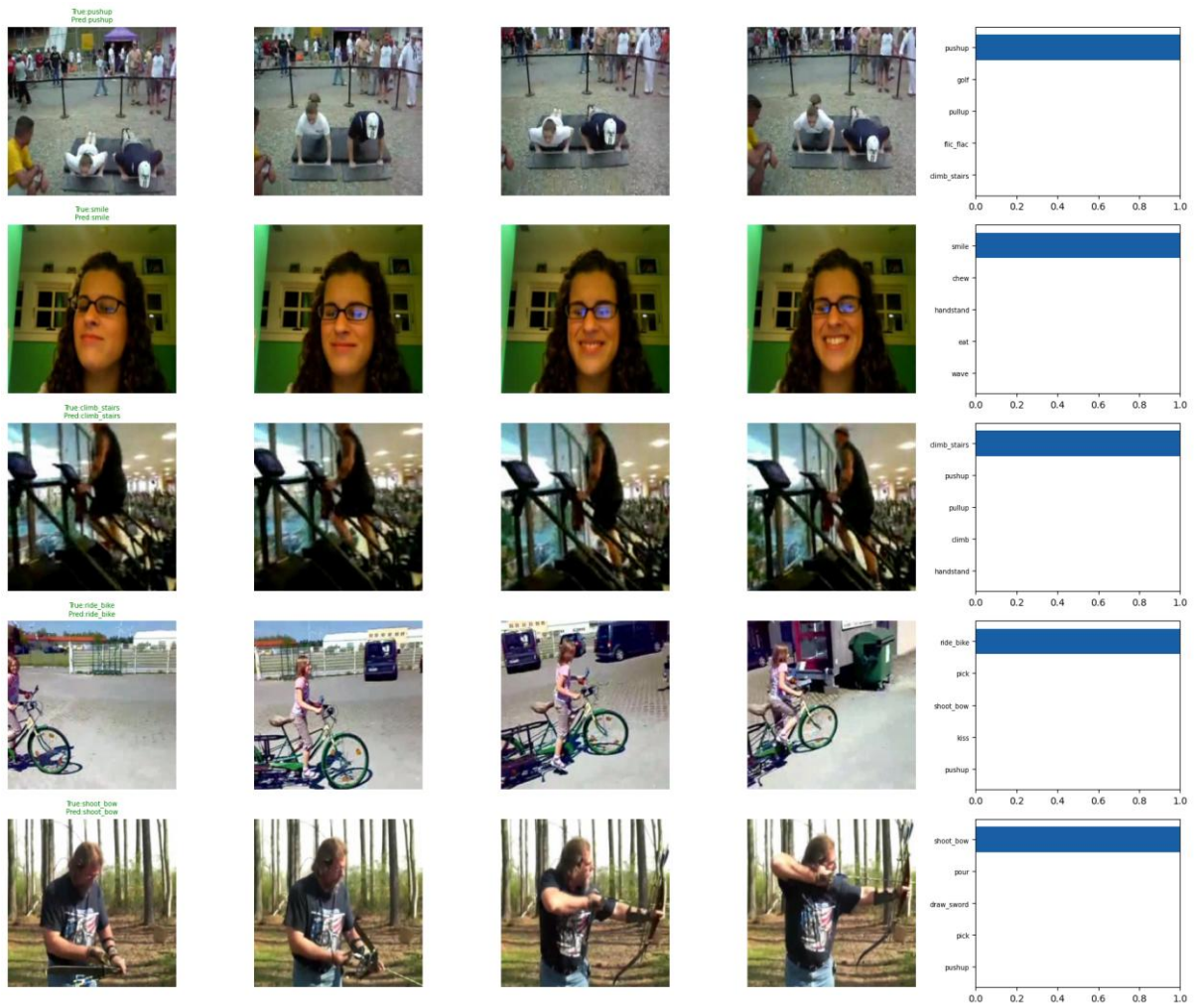


Fig. 9. Error analysis: correctly classified HMDB\_simp test examples with top-5 confidence scores.



Figure 10

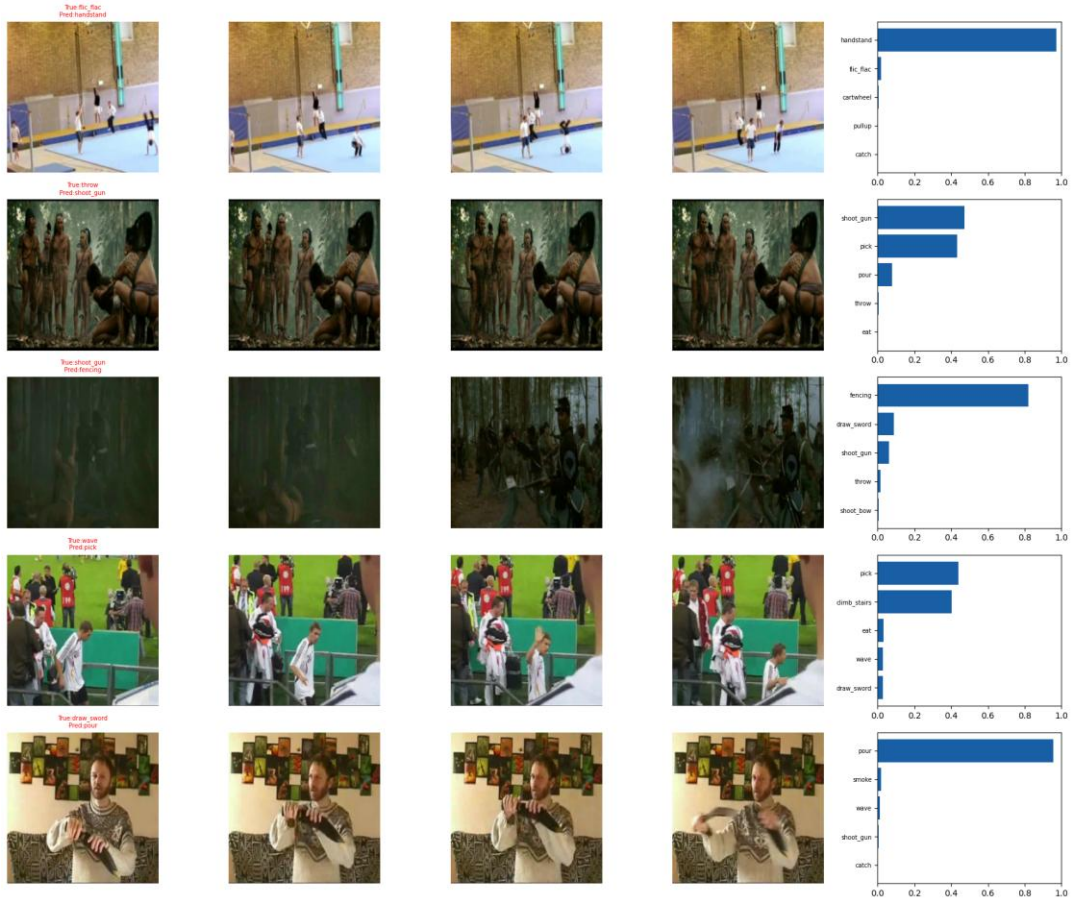
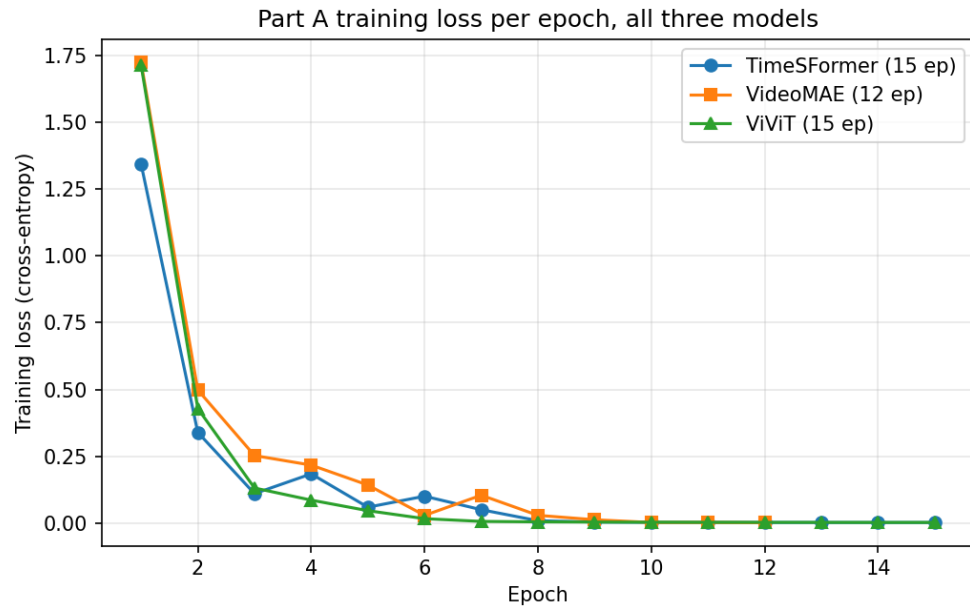


Fig 10: Error analysis: misclassified HMDB\_simp test examples with top-5 confidence scores. All five failures involve semantically or visually similar class pairs (e.g. `flic_flac`→`handstand`, `wave`→`pick`), consistent with the confusion-matrix patterns in Fig. 2.

Two distinct failure modes are visible above. Three errors stem from genuine visual/pose ambiguity: `flic_flac` is predicted as `handstand` in a gymnastics scene where both actions share an inverted-body pose at the sampled frames; `wave` is predicted as `pick` in a crowd scene where an arm-raising gesture is visually compatible with either label (the model splits confidence almost evenly, 0.42 vs. 0.40); and `draw_sword` is predicted as `pour` with high confidence (0.85) because the two-handed grip of an object near the body looks similar for both actions in this clip. The remaining two errors are better explained by scene conditions than by motion: `shoot_gun` is predicted as `fencing` with high confidence (0.80) in a very low-light forest scene where the actor is barely visible in three of four sampled frames, and `throw` is predicted as `shoot_gun` in a dark, cluttered scene where confidence is split across three classes (`shoot_gun` 0.45, `pick` 0.40, `pour` 0.12), indicating genuine model uncertainty rather than a confident wrong association. This suggests VideoMAE's masked-reconstruction features are more robust to pose ambiguity than to poor lighting/occlusion, since the confidently-wrong predictions cluster in well-lit, clearly-visible clips while the low-visibility clips instead produce spread-out, low-confidence predictions.

**Figure 11**



*Fig. 11. Training loss per epoch for all three Part A models. Note: the main training loop logs per-epoch training loss and test-set accuracy only at milestone epochs; a separate held-out validation split with its own loss/accuracy curve was tracked only for the 5-epoch ablation runs in Part B, not for the full Part A training, so no validation-loss curve is shown here.*

**Table 3**

| <b>Class</b>    | <b>Precision</b> | <b>Recall</b> | <b>F1</b>    |
|-----------------|------------------|---------------|--------------|
| brush_hair      | 0.875            | 1.000         | 0.933        |
| cartwheel       | 0.833            | 0.714         | 0.769        |
| catch           | 1.000            | 1.000         | 1.000        |
| chew            | 1.000            | 1.000         | 1.000        |
| climb           | 0.889            | 1.000         | 0.941        |
| climb_stairs    | 0.875            | 1.000         | 0.933        |
| draw_sword      | 1.000            | 0.750         | 0.857        |
| eat             | 0.778            | 0.875         | 0.824        |
| fencing         | 0.857            | 0.857         | 0.857        |
| flic_flac       | 0.714            | 0.714         | 0.714        |
| golf            | 0.889            | 1.000         | 0.941        |
| handstand       | 0.875            | 1.000         | 0.933        |
| kiss            | 1.000            | 1.000         | 1.000        |
| pick            | 0.857            | 0.857         | 0.857        |
| pour            | 0.800            | 1.000         | 0.889        |
| pullup          | 1.000            | 1.000         | 1.000        |
| pushup          | 1.000            | 1.000         | 1.000        |
| ride_bike       | 1.000            | 0.875         | 0.933        |
| shoot_bow       | 0.889            | 1.000         | 0.941        |
| shoot_gun       | 0.714            | 0.625         | 0.667        |
| situp           | 1.000            | 1.000         | 1.000        |
| smile           | 1.000            | 1.000         | 1.000        |
| smoke           | 1.000            | 0.714         | 0.833        |
| throw           | 1.000            | 0.571         | 0.727        |
| wave            | 0.778            | 0.875         | 0.824        |
| <b>microavg</b> | <b>0.895</b>     | <b>0.900</b>  | <b>0.895</b> |

Table 4: Full 21 Class [Frame-AP@0.5](#) Breakdown on JHMDB test set.

| Class          | Frame-AP |
|----------------|----------|
| pour           | 0.899    |
| pullup         | 0.836    |
| clap           | 0.545    |
| shoot_bow      | 0.540    |
| swing_baseball | 0.413    |
| brush_hair     | 0.389    |
| shoot_gun      | 0.292    |
| push           | 0.237    |
| golf           | 0.218    |
| wave           | 0.159    |
| climb_stairs   | 0.152    |
| throw          | 0.130    |
| kick_ball      | 0.120    |
| stand          | 0.102    |
| run            | 0.088    |
| walk           | 0.071    |
| jump           | 0.030    |
| shoot_ball     | 0.025    |
| sit            | 0.007    |
| pick           | 0.003    |
| catch          | 0.001    |